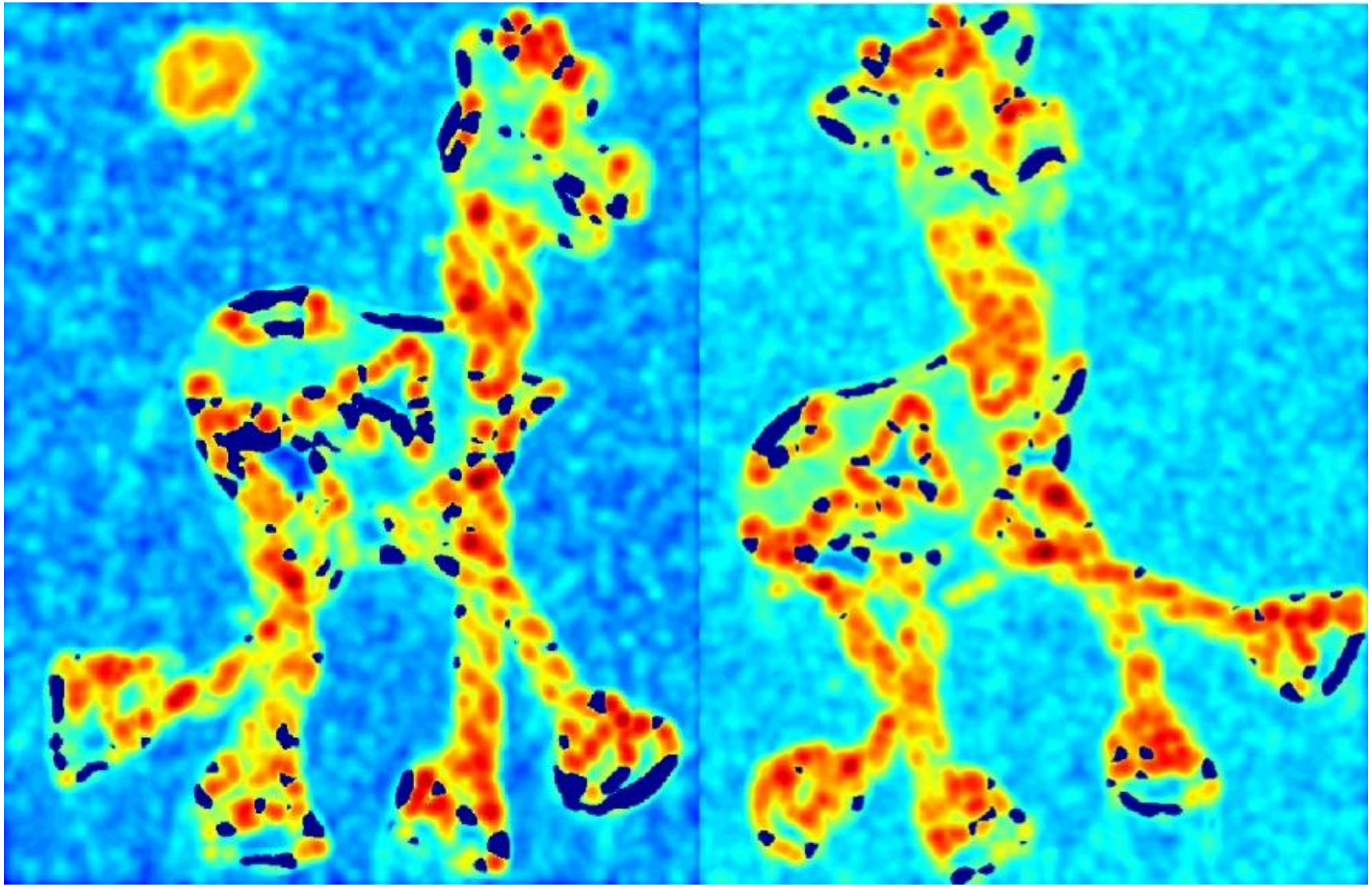
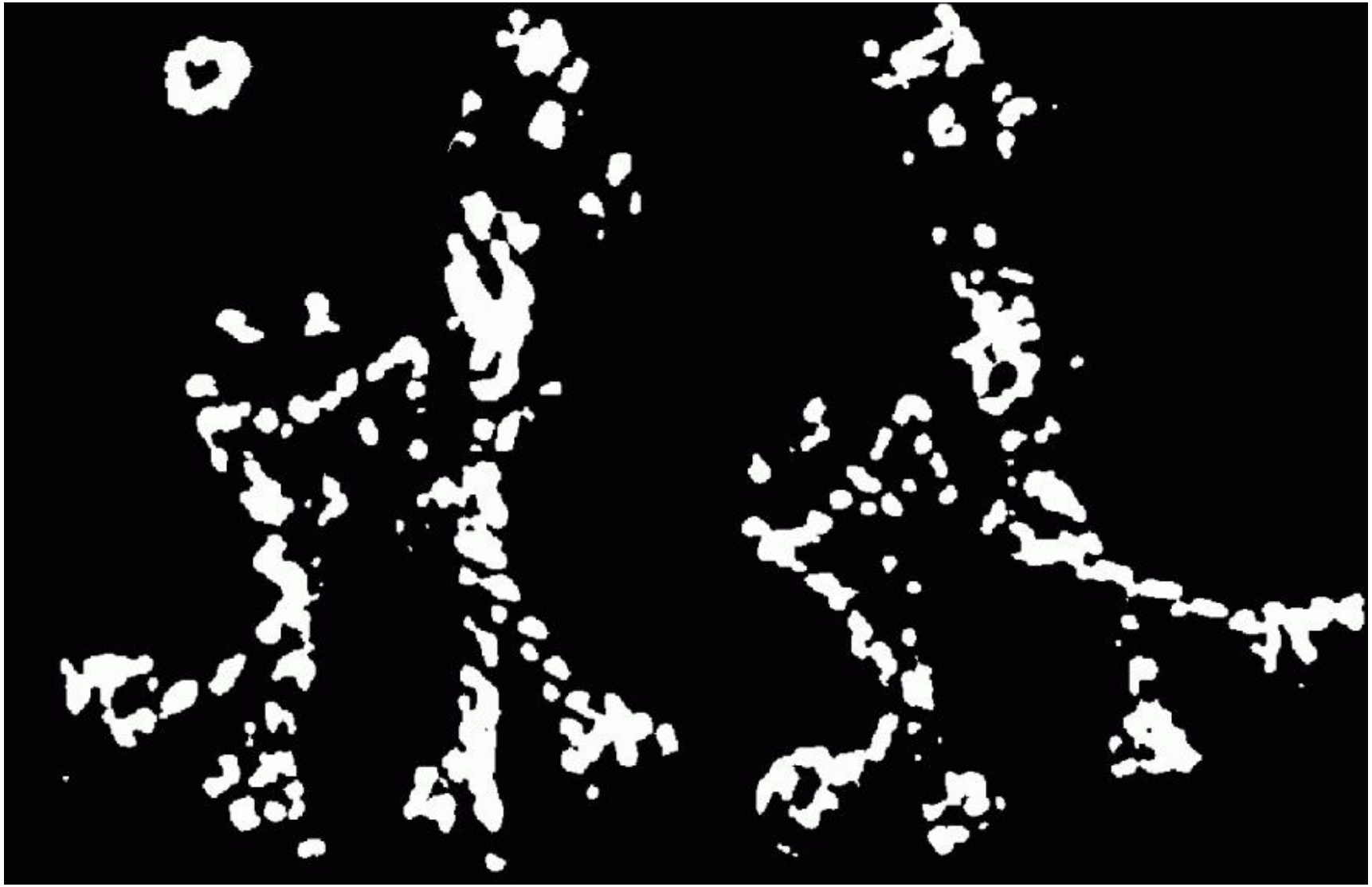
Harris detector example



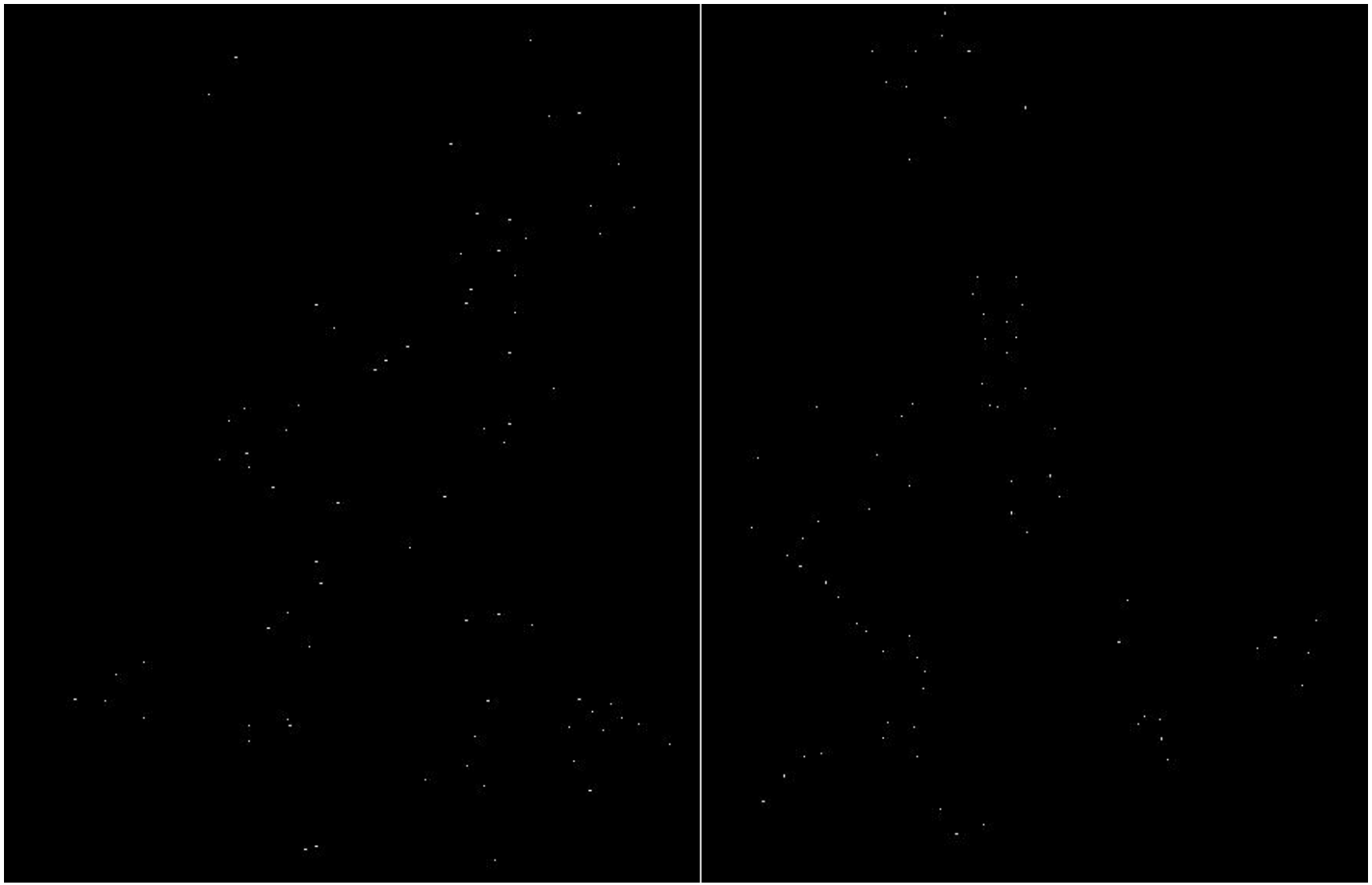
f value (red high, blue low)



Threshold (f > value)



Find local maxima of f (non-max suppression)



Harris features (in red)



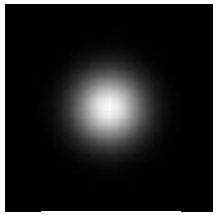
Weighting the derivatives

- In practice, using a simple window *W* doesn't work too well

$$H = \sum_{(x,y)\in W} \begin{bmatrix} I_x^2 & I_x I_y \\ I_x I_y & I_y^2 \end{bmatrix}$$

- Instead, we'll *weight* each derivative value based on its distance from the center pixel

$$H = \sum_{(x,y)\in W} w_{x,y} \begin{bmatrix} I_x^2 & I_x I_y \\ I_x I_y & I_y^2 \end{bmatrix}$$



$w_{x,y}$

Harris Detector – Recap [Harris88]

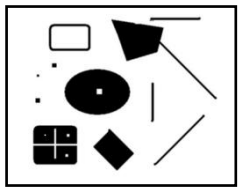
Second moment matrix

$$\mu(\sigma_I, \sigma_D) = g(\sigma_I) * \begin{bmatrix} I_x^2(\sigma_D) & I_x I_y(\sigma_D) \\ I_x I_y(\sigma_D) & I_y^2(\sigma_D) \end{bmatrix}$$

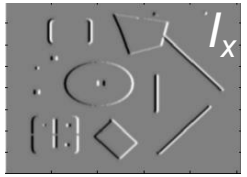
$$\det M = \lambda_1 \lambda_2$$

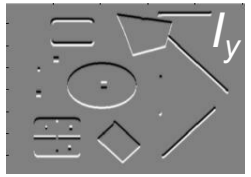
$$\text{trace } M = \lambda_1 + \lambda_2$$

0. Input image



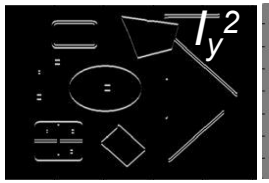
1. Image derivatives

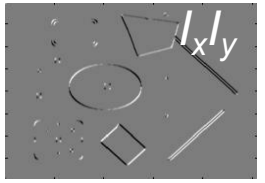




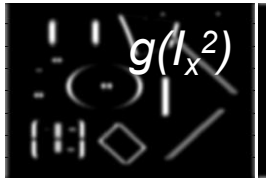
2. Square of derivatives



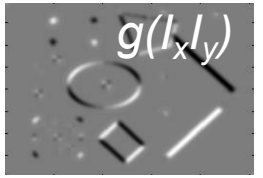




3. Gaussian filter $g(s_I)$







4. Cornerness function – both eigenvalues are strong



5. Non-maxima suppression

Harris Corners – Why so complicated?

- Can’t we just check for regions with lots of gradients in the x and y directions?
  - No! A diagonal line would satisfy that criteria

